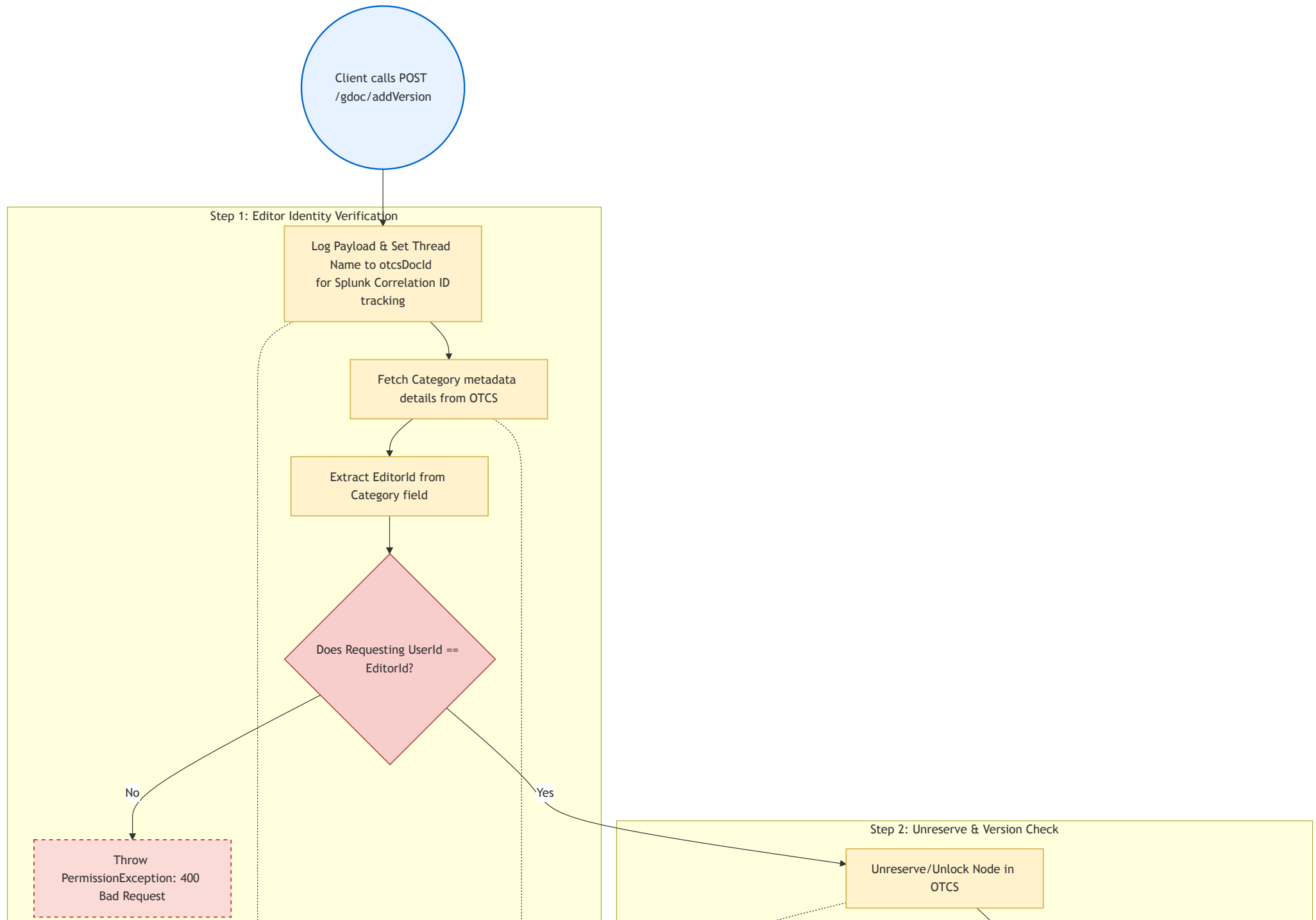


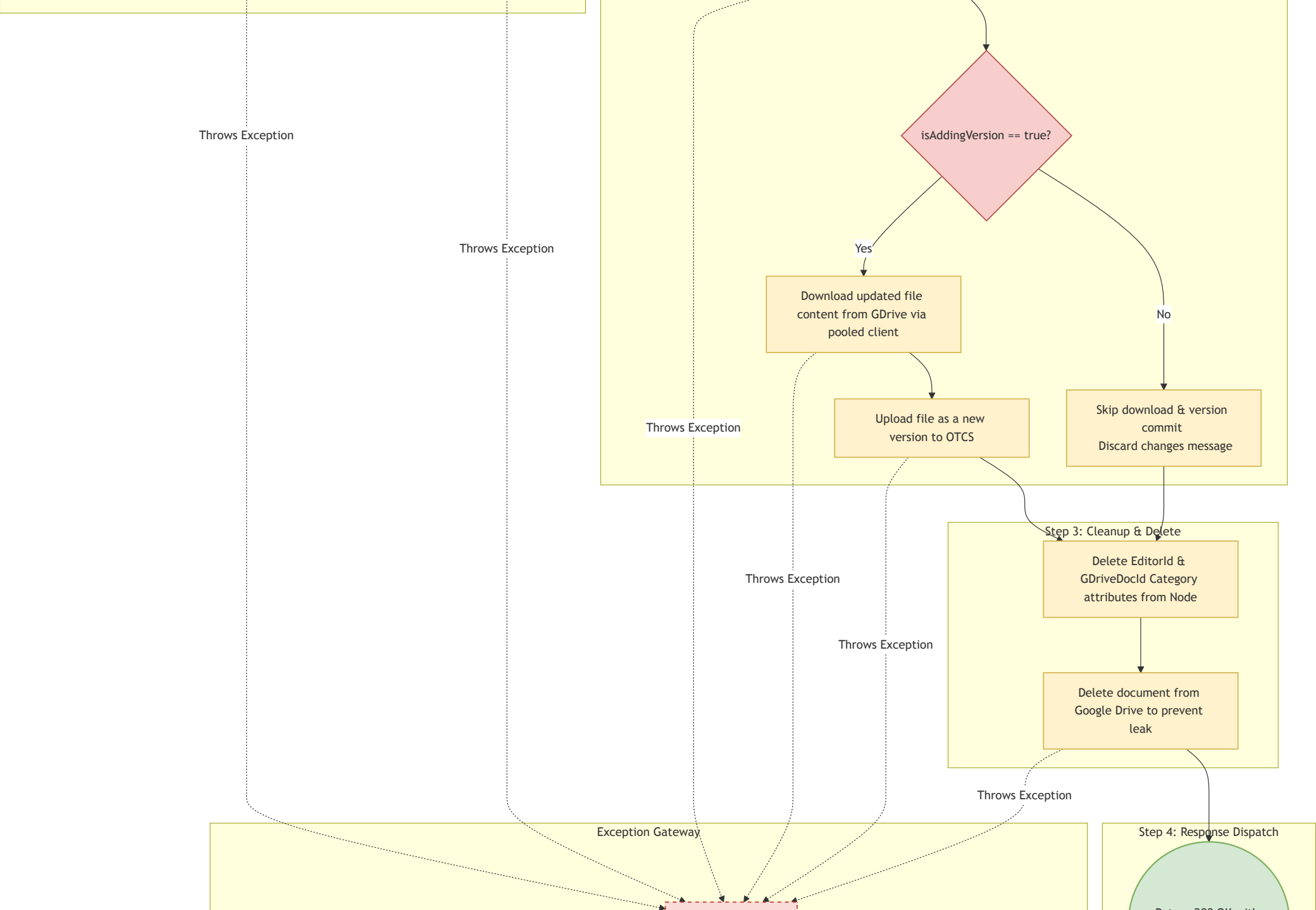
Operational Flow & Exception Handling: /gdoc/addVersion

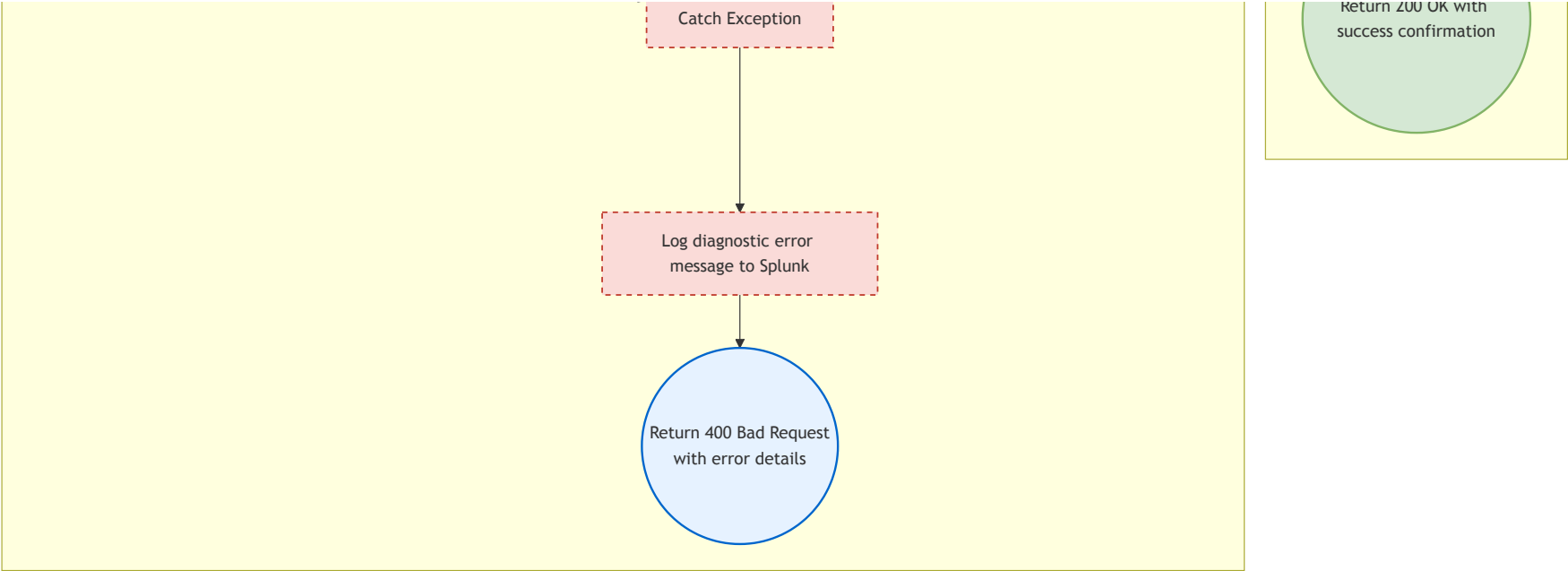
This document details the complete end-to-end execution flow and exception propagation system for the /gdoc/addVersion endpoint.

1. Process Flow Diagram (Boxes & Arrows)

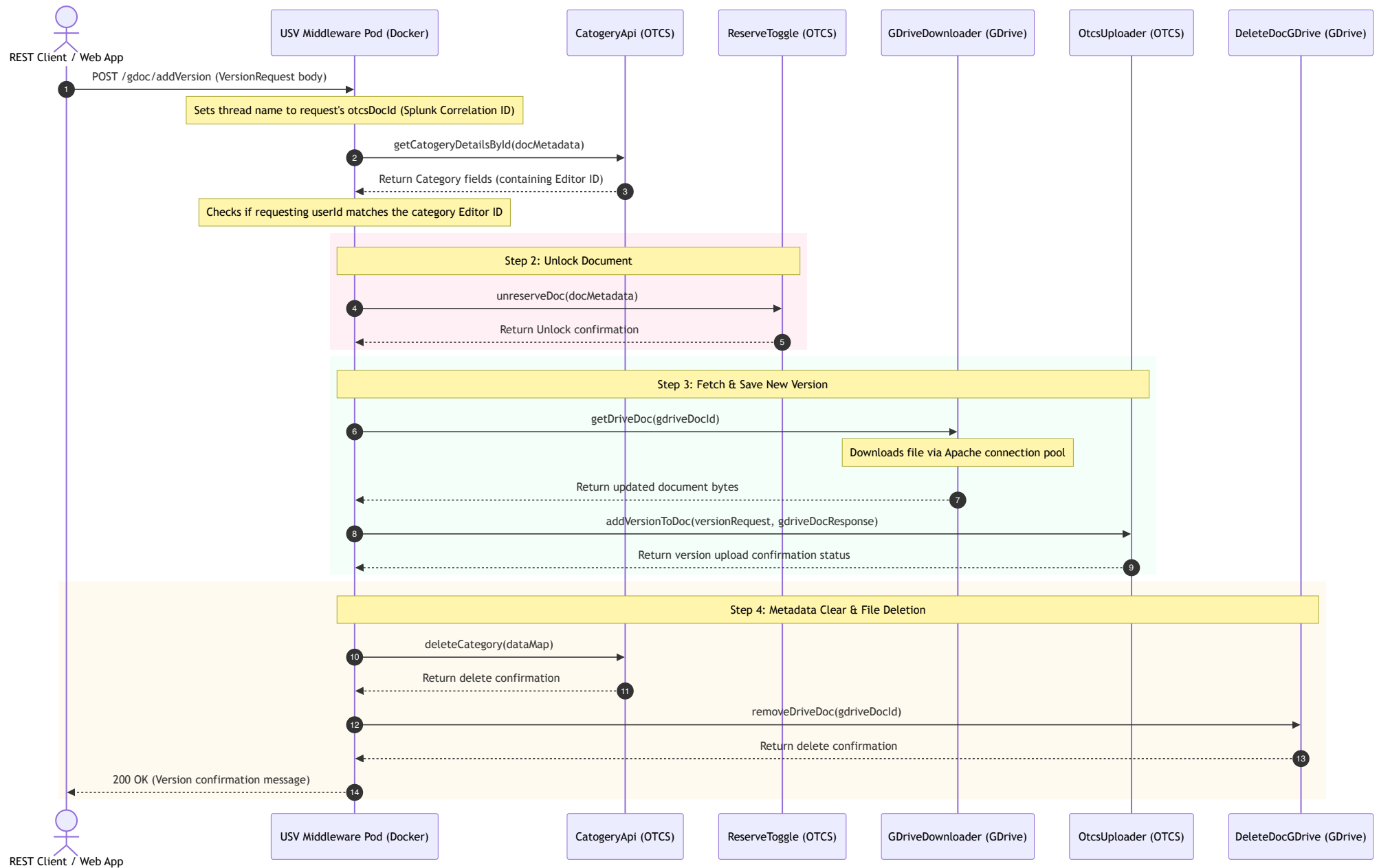
This flowchart traces the step-by-step process of the /gdoc/addVersion endpoint, highlighting decision gates (diamonds), logical operations (rectangles), and the execution path.







2. Happy Path Sequence Diagram



3. Step-by-Step Execution Mechanics

1. Entry Point (`ReqController.java#addVersion`):

- Listens on `POST` requests at `/gdoc/addVersion`.
- Modifies the execution thread name temporarily to `request.getOtcDocId()` to track metrics (Correlation ID in Splunk/ELK).
- Invokes `usvService.addVersionToDoc(request)`.

2. Verify Editor ID (`UsvService.java#addVersionToDocument`):

- Queries the document metadata from OTCS via `catogeryApi.getCatogeryDetailsById` to read the `EditorId` attribute.
- Evaluates the Editor ID against the requesting user ID.
- If they do **not** match, it blocks execution, logs a warning, and returns a `400 Bad Request` containing a `PermissionException` message.

3. Unlock Document (`UsvService.java#addVersionToDocument`):

- Releases the reservation lock on Content Server by executing `reserveToggle.unreserveDoc`.

4. Add Version or Discard Changes:

- **Scenario A: `addingVersion` is True (Check-in edits):**
 - Downloads the modified document version from Google Drive via `gdriveDownloader.getDriveDoc` (making a pooled HTTP connection).
 - Commits the downloaded document bytes as a new version on Content Server via `otcsUploader.addVersionToDoc`.
- **Scenario B: `addingVersion` is False (Discard edits):**
 - Skips the Google download and version upload.
 - Logs a discard action and prepares a discard confirmation.

5. Metadata Cleanup & Google File Deletion:

- Invokes `categoryApi.deleteCategory` to clear the `EditorId` and `GDriveDocId` category fields from the OTCS node.
- Invokes `deleteDocGDrive.removeDriveDoc` to delete the temporary document from Google Drive to satisfy data residency requirements.
- Returns the confirmation response.

4. Exception Handling & Propagation Details

Downstream Error Translation

1. Downstream API Failures:

- HTTP clients capture non-2xx status codes (such as Google Drive access token expiration, network timeouts, or OTCS check-in failures).

2. Controller Level Try-Catch Wrapper:

- `UsvService.java#addVersionToDoc` handles execution errors within a `try-catch` block.
- If any downstream API error occurs, the code catches the exception and returns the error message directly inside a `ResponseEntity` with a status of `400 Bad Request`.